

Assistive Reading & Tutoring System

Technical Specification v3 — Three-Feature Demo Scope | Tech4City 2026 | Supersedes the Pi-based spec v1

1. Problem Statement

Students with dyslexia, ADHD, and other learning differences are granted human reader support under MOE's Access Arrangements for national exams such as the PSLE. An approved reader reads Math and Science exam questions aloud exactly as written, without explaining or summarising; this accommodation is not provided for Language papers, since reading comprehension is what those papers assess. The same support does not extend to daily homework: schools lack staffing for one-on-one reading help outside exam periods, and parents are often unavailable or untrained to act as a neutral, non-explaining reader under the same standard. As a result, an estimated 28,400 mainstream-school students with mild to moderate learning needs in Singapore study for their most consequential exam without the accommodation they are officially entitled to during it.

An accommodation that only reads *for* a child builds dependency without building skill. Clinical guidance on dyslexia is that meaningful reading gains require structured, explicit practice with multisensory (visual, auditory, tactile) elements. The system therefore treats a stuck word not only as something to read aloud, but as a teaching moment — while giving parents visibility into study patterns they otherwise never see.

2. Solution Overview

A mobile app is the product; the only hardware is furniture. The family's phone sits in an ordinary commercial folding stand, screen facing the student. A small mirror clip over the front camera redirects the camera's view downward onto the worksheet lying flat in front of the stand — the proven optical pattern of the Osmo Reflector for iPad education systems. The student sees the screen, works on paper in their natural position, and points with a free hand. No single-board computer, no laser, no servos, no custom wireless links.



Physical configuration: phone in a commercial stand, mirror clip over the front camera, camera view path redirected down to the worksheet zone in front of the stand, screen facing the student.

Scope discipline: a live demo runs 2–3 minutes regardless of build size, so v3 ships exactly three features — each one demonstrable, each one covering a different judging axis:

- **Exam-Prep Mode** — SEAB-standard literal reading with built-in session analytics, exportable reports, and a Telegram parent loop. (Feasibility + Social Impact)
- **AI Tutoring** — vision-LLM step-by-step explanation with on-screen highlight sync. (Creativity + the capability a reader pen structurally cannot offer)
- **Stuck-Word Autopsy + Trace-to-Unlock** — the phonics moment: grapheme-chunk sound-out and camera-verified finger tracing on real paper. (Creativity + the fading-support answer to the dependency question)

3. What Changed From Spec v1 (Pi-Based Architecture)

v1 built a standalone hardware device: a Raspberry Pi Zero 2 W with a Pi Camera Module 3 on a lamp-style stand, a laser pointer on a 2-axis servo bracket (PCA9685 driver) for physical highlighting, a Pi-hosted WiFi access point, and a local Flask/FastAPI server streaming audio to a phone acting as a thin peripheral. v3 deletes that architecture and adds two capabilities v1 never had. Component by component:

v1	v3	Why
Raspberry Pi Zero 2 W + Pi Camera Module 3 on a custom lamp-style stand	The family's phone in a commercial folding stand; front camera + mirror clip (Osmo Reflector pattern) + mandatory frame-flip (pipeline step 0)	The phone camera is better hardware than the Pi camera and is already owned. Removes an entire compute board, its power, SD card, and OS. The mirror restores the near-top-down view while keeping the screen visible — the one thing v1's overhead camera could never offer.
Laser diode on a 2-axis servo bracket + PCA9685 PWM driver	On-screen highlighting only	Servo resolution ($\sim 1^\circ \approx 6\text{mm}$ at working distance) is the same size as a grapheme target (4–8mm), so the laser physically cannot do fine highlighting. The screen is pixel-precise for free, and the child-eye-safety burden disappears entirely.
Pi-hosted WiFi access point (hostapd) + local Flask/FastAPI server + phone-as-peripheral audio streaming	Nothing — the app runs where the mic, speaker, screen, and camera already are	This was v1's largest untested subsystem and biggest live-demo failure surface. It existed only to bridge the Pi and the phone; with no Pi, the bridge has no job.
Session ends when the reading ends — no after-session value	Session analytics page (requests, re-reads, pacing, most-requested words), PDF + XLSX exports, and a Telegram parent loop with LLM-generated date-range reviews	Turns a reading tool into parent visibility. Logging is event-based (timestamps only); the mic serves trigger detection and no raw audio is ever retained.
Accommodation only — the device reads, forever	Stuck-Word Autopsy + Trace-to-Unlock: grapheme-chunk sound-out from a static phoneme bank, then camera-verified finger tracing on the real paper	The direct answer to “aren't you replacing reading practice with listening?” Multisensory phonics on real homework, zero extra hardware.
“Practice Mode”	Renamed “AI Tutoring” — behaviour unchanged	Clearer to judges and parents.

v1	v3	Why
Unchanged from v1	Deterministic Exam-Prep reading path (no LLM anywhere in it), Huawei Cloud OCR, Azure Speech WordBoundary highlight sync, DeepSeek/GLM via Huawei Cloud MaaS for reasoning	The compliance guarantee and the cloud stack were right the first time; only the delivery hardware changed.

4. Physical Setup & Pipeline Step Zero

- 1 Phone is placed upright in the folding stand, screen facing the student.
- 2 The mirror clip attaches to the phone's top edge, angled ~45° over the front camera, bending its view 90° downward.
- 3 The worksheet lies flat on the desk directly in front of the stand — between the student and the phone, in the child's natural working position.
- 4 Optional LED ring clips to the stand for even worksheet lighting (cheap OCR-accuracy win; not load-bearing).
- 5 **Step 0 of every pipeline:** a single mirror reflection produces a mirror-flipped image, so every captured frame is flipped along one axis (cv2.flip / canvas scale(-1,1)) before OCR, MediaPipe, or display sees it. The flip axis is fixed by the clip's orientation — determined once during the first physical test and hardcoded.

Known empirical unknown (test in week one): mirror clip height and angle determine the visible page area, and clip fit varies with each phone model's front-camera position. Prototype with a low-cost craft mirror and tape on the team's demo phone before purchasing a clip.

5. Features

5.1 Feature 1 — Exam-Prep Mode (literal reading + session analytics)

During the session:

- 1 Student enters Exam-Prep Mode. The system announces aloud that session logging has started (spoken cue; no persistent recording badge on the UI). Logging runs for the whole session.
- 2 Student points at a word or question and says the trigger phrase (“read this”).
- 3 The app keyword-matches the trigger (no LLM), captures a single frame, and flips it (step 0).
- 4 MediaPipe Hands locates the fingertip; Huawei Cloud OCR returns text + per-word bounding boxes; the nearest box or box group to the fingertip is selected.
- 5 Azure Speech reads the selected text verbatim; on-screen karaoke highlighting syncs via native WordBoundary events. Nothing is explained, defined, or paraphrased — no reasoning model exists anywhere in this path.
- 6 Every interaction writes a timestamped event to the session log: request time, selected words, question region, re-read or first read. The mic is live only for trigger detection; no raw audio is retained.

At session end — the analytics page:

- An interactive statistics page renders in the app: total requests and re-reads, re-read clustering by question type, most-requested words, a pacing timeline across the session (including slowdown patterns as a fatigue signal), and per-question time cost of reading requests.
- Insights are labelled **struggle & engagement indicators** — derived purely from events and timing. The system makes no emotional or clinical claims.
- Manual downloads: a PDF report (charts + written insights) and an XLSX workbook (all raw statistics).

The Telegram parent loop:

- On session end, the bot automatically sends both files to the configured destination: the family group chat (bot added to the group) or the parent directly. **Constraint:** Telegram bots cannot initiate a chat — for direct delivery the parent must tap /start on the bot once during setup; a user_id from @userinfobot alone is not sufficient.
- The parent can query the bot through buttons and date inputs: a specific date's report or a monthly rollup.
- For any date-range request, the bot sends the stored event summaries for that range to the reasoning model (Huawei Cloud MaaS) and returns a narrative report: trends, study-pattern insights, and recommendations — framed strictly as study patterns, never as emotional or clinical assessment.

5.2 Feature 2 — AI Tutoring

- 1 Student asks a question by voice or text (text input always available as the Safari/iOS fallback).
- 2 The app captures and flips a frame, freezes it on screen, and sends frame + question to Huawei Cloud MaaS (DeepSeek-V3.1 / GLM-5).
- 3 The model returns a step-by-step explanation with a target region per step.
- 4 Each step's text goes to Azure Speech; the app narrates while highlighting that step's region on the frozen frame.
- 5 Steps advance one at a time; the student can ask follow-ups.

5.3 Feature 3 — Stuck-Word Autopsy + Trace-to-Unlock

- 1 During reading, the student taps (or points at) a stuck word: the app speaks that word only, then goes silent. The word and its grapheme pattern are logged.
- 2 A second tap triggers “sound it out”: the app splits the word's OCR bounding box proportionally into grapheme chunks (ch–ar–ge).
- 3 The on-screen highlight sweeps chunk by chunk, playing each phoneme from the static audio bank, then blends the whole word at normal speed via TTS.
- 4 The app prompts “now trace it.” The student traces the word on the physical paper with a finger.
- 5 MediaPipe verifies loosely at reduced frame rate (fingertip inside the word box, net left-to-right motion); phonemes play as the finger passes each chunk.
- 6 On completion, a chime sounds and the word unlocks. Visual + auditory + tactile in one action, on real homework, with zero extra hardware.

Phoneme audio bank: a static bank of the ~44 English phoneme sounds, sourced from openly-licensed recordings (e.g. CC-licensed phoneme audio hosted on Wikimedia Commons, as used on Wikipedia's IPA pages), with self-recorded gap-fill only where a licensed file is missing or poor. Each file's license must be verified and attributed.

TTS is never used for isolated phonemes: neural TTS hallucinates a trailing schwa on isolated plosives (“buh” instead of /b/), which disrupts blending and is pedagogically harmful for dyslexic learners. Programmatic syllable-splicing was evaluated and rejected as non-deterministic and brittle. A static bank is zero-latency, offline-capable, and acoustically identical on every trigger.

6. Tech Stack

Layer	Technology
Product	Mobile-first web app (React or similar); the phone provides camera, screen, mic, speaker, and compute
Physical	Commercial folding phone stand (user-provided); mirror clip over the front camera (Osmo Reflector pattern); optional clip-on LED ring
Frame preprocessing	Mandatory mirror-flip on every frame (pipeline step 0); axis hardcoded after the first physical test
Hand / finger detection	MediaPipe Hands — single-shot for pointing; reduced-frame-rate loose verification for tracing
OCR (deterministic paths)	Huawei Cloud OCR (REST API), text + per-word bounding boxes
Reasoning (AI Tutoring + bot reviews only)	DeepSeek-V3.1 / GLM-5 via Huawei Cloud MaaS (Anthropic-/OpenAI-compatible endpoint); region availability to be re-verified in the console
Text-to-speech	Microsoft Azure Speech (official SDK), native WordBoundary events for highlight sync
Phoneme audio	Static bank of ~44 openly-licensed phoneme recordings (license-checked, attributed); self-recorded gap-fill; never TTS
Session data	Timestamped interaction event log per session; no raw audio retention; drives the analytics page, exports, and bot reviews
Report generation	PDF: charts + insights (client-side jsPDF/Chart.js render or server-side ReportLab). XLSX: SheetJS (client) or openpyxl (server)
Parent loop	Telegram Bot API (polling mode); delivery to family group chat or /start-onboarded parent DM; button + date-input queries; LLM-generated range reviews
Backend	Thin key-proxy server for Huawei OCR, Huawei MaaS, and Azure Speech calls (API keys never ship in client code); stores session event summaries for bot queries

7. Key Design Principles

- **No reasoning model in any compliance-critical reading path.** Exam-Prep reading and the phoneme sound-out are fully deterministic (OCR + index matching + static audio). The literal-reading guarantee is a property of the architecture, not a prompt.
- **Indicators, not diagnosis.** All analytics derive from interaction events and timing. The system never claims to read a child's emotions and never uses clinical language. Bot reviews output study-pattern recommendations only.

- **No raw audio retention.** The microphone serves trigger detection and question capture; nothing else is stored. The session log is events and timestamps.
- **Support fades by design.** Tap-a-word reads one word, not the page; the Autopsy teaches the pattern behind the word; tracing rehearses it. The accommodation is built to become less necessary.
- **Every physical component earns its place.** The stand gives fixed geometry, the mirror gives a near-top-down view while keeping the screen visible, the LED ring gives OCR reliability. Nothing is decorative.
- **Attention cues per mode.** Eyes on the screen during sound-out, on the paper during tracing; each transition is announced aloud to prevent split-attention confusion, particularly for ADHD users.
- **Never invent pedagogy.** Grapheme ordering and phonics behaviour anchor to an established, published phonics scope-and-sequence.

8. Open Items

- Week-one mirror prototype: craft mirror + tape on the demo phone; hardcode the flip axis; then buy the matching clip.
- Curate the phoneme bank: download candidate files, verify each license, record attributions, self-record gaps.
- Verify Huawei Cloud MaaS region availability (AP-Singapore vs CN-Hong Kong) in the console before any data-residency claim.
- Telegram onboarding flow: bot-to-group setup and the one-time parent /start step for direct delivery.
- Contact DAS: pilot interest, and whether exported session reports are useful supporting material for Access Arrangements applications.
- Test Web Speech API voice input on Safari/iOS; the text-input fallback stays permanently available.